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Luca Podavini

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ABOUT MYSELF

Master's student in Artificial Intelligence Systems at the University of Trento, specializing in Intelligent Robots. Building on a solid foundation in software and backend development, I am developing advanced skills in AI, such as Reinforcement Learning, Computer Vision, and Language Models, applied to the field of robotics.

Please visit my website (lucapodavini.it) for a more interactive and easy to read bio. You will also find the complete project portfolio with references to the technical reports, GitHub repositories and demos.

The Italian version of this document is available at www.lucapodavini.it/documents/luca-podavini-cv-it.pdf

SKILLS

C/C++ | Rust | Python | PyTorch | ROS/ROS 2 | Robot Planning | Reactive Control | Optimal Control | Deep Learning | Reinforcement Learning | Computer Vision | Embedded Software | Docker | Git

WORK EXPERIENCE

30/11/2019 - 31/08/2022 - PUEGNAGO DEL GARDA, ITALY

PROGRAMMERCUALEVA S.R.L

Rapidly transitioned from an internship to a key part-time role, demonstrating strong technical aptitude and commitment during my final year of high school. Engineered and maintained core functionalities for enterprise-grade web, desktop, and mobile solutions, leveraging the .NET Framework (C#).

09/2019 - 09/2019 - PRAGUE, CZECHIA

PROGRAMMERTHINK EASY S.R.O

Gained hands-on experience in back-end web development using Django (Python) and front-end development using React (JS) during a one-month internship abroad.

EDUCATION & TRAINING

31/08/2024 - Current - TRENTO, ITALY

MASTER DEGREE IN ARTIFICIAL INTELLIGENCE SYSTEMS - UNIVERSITY OF TRENTO

Field(s) of study: Intelligent Robots | **Level in EQF:** 7 | **Website:** www.unitn.it

31/08/2021 - 17/09/2024 - TRENTO, ITALY

BACHELOR'S DEGREE IN COMPUTER SCIENCE- UNIVERSITY OF TRENTO

Field(s) of study: Information and Communication Technologies | **Final grade:** 98/110 | **Level in EQF:** 6 | **Thesis:** Move Fuzzer | **Website:** www.unitn.it/

31/08/2016 - 26/06/2021 - LONATO DEL GARDA, ITALY

DIPLOMA IN INFORMATION AND COMMUNICATION TECHNOLOGIES- I.I.S LUIGI CEREBOTANI

Field(s) of study: Information and Communication Technologies | **Final grade:** 100/100 | **Level in EQF:** 4 | **Website:** https://iislonato.edu.it/

PROJECTS

30/09/2025 - 31/12/2025

REINFORCEMENT LEARNING for UNICYCLE ROBOT NAVIGATION Final and most significant assignment of the Optimization and Learning for Robot Control course. Implemented the **PPO** (Proximal Policy Optimization) **Reinforcement Learning** algorithm to solve a full autonomous navigation problem in an obstacle-dense environment.
Link www.lucapodavini.it/#unicycle-rl

31/12/2025 - 31/01/2026

Robot Motion Planning - Victim Rescue Motion planning project on an Agile X LIMO. Explored both **Combinatorial Planning** (Exact/Approximate Cell Decomposition) and **Sampling Based Planning** (PRM), to solve the Victim Rescue problem in a controlled environment with ROS.
Link www.lucapodavini.it/#robot-planning

30/04/2025 - 30/06/2025

Deep Learning: Parameter Efficient Fine Tuning for CLIP This project explores Parameter Efficient Fine Tuning (PEFT) techniques applied to the **CLIP** model for image classification tasks. We implemented and compared several PEFT methods, including CoCoOp, **CLIP-LoRA**, and **DISEFT**, to evaluate their performance in few-shot learning scenarios.
Link www.lucapodavini.it/#deep-learning

31/03/2025 - 31/05/2025

Natural Language Understanding This project focuses on **Language Modelling (LM)** and **Natural Language Understanding (NLU)**. Various architectures have been compared, including **LSTM**-based models with dropout layers and advanced optimization techniques like AdamW and NT-AvSGD, as well as transformer-based models such as **BERT**.
Link www.lucapodavini.it/#nlu

30/09/2023 - 31/03/2024

CLAW MACHINE Designed and built an IoT claw machine from scratch, including custom aluminum framing, 3D printed parts, electronics, and full Embedded C software architecture.

This project was presented at Fablab Trento during **ICT Days 2024**.
Link www.lucapodavini.it/#claw-machine

31/07/2025 - CURRENT

VISUAL-HIVE VJ software developed as a personal project to meet the specific needs of live events, prioritizing simplicity and real-time performance. It enables seamless overlaying of foreground logos onto dynamic background loops, with integrated effects like strobe and beat-synced motion.

The project features advanced**real time** data processing, multi-threading, **MIDI** controls,**Ableton Link** integration and real time BPM detection.
Link www.lucapodavini.it/#visual-hive

LANGUAGE SKILLS

Mother tongue(s): **ITALIAN**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	B2	B2	B2	B2	B2